SHOC SOPs 2

Early Warning and Monitoring SOPs

Multi hazard detection, analysis, and anticipatory action capability.

SADC Humanitarian Emergency Operations Centre, Nacala, Mozambique

STANDARD OPERATING PROCEDURES

EARLY WARNING AND MONITORING SYSTEMS

BENCHMARKED AGAINST:

IFRC Surge Operational Toolbox 2024 | UNDAC Handbook 8th Ed. | WMO CAP 1.2 (reference)
SADC MHEWS Framework | ISO 22301:2019 | ISO 27001:2022 | OCHA AA Framework

Version 3.3 — 08 March 2026

Effective Date: 01 April 2026

Table of Contents

Table of Contents 3
1. Introduction 6
1.1 Mandate Clarification 6
1.2 SHOC Institutional Identity (per MoA) 6
1.3 SHOC Staff Designations (per MoA) 6
1.4 Legal and Institutional Basis 7
1.5 Guiding Principles 7
1.6 Reference Standards and Technical Frameworks 8
1.7 Information Sourcing Hierarchy 8
Primary Technical Sources 8
Continental and Regional Economic Community MHEWS Mechanisms 8
Secondary and Supporting Sources 9
Note: CAP and AMHEWAS Alignment 9
1.8 Information Conflict Resolution Protocol 9
2. SHOC Operational Framework for Early Warning and Monitoring 10
2.1 Operational Phase Definitions 10
2.2 SHOC Operational Phase Framework 10
2.3 Decision Authority Matrix 10
2.4 Cross-SOP Integration 11
2.4.1 External Integration (Other SOPs) 11
2.4.2 Internal Integration (SOPs 2.1–2.4 Handoff Points) 11
3. SHOC Situation Room Operations and Mapping Systems 12
3.1 Situation Room Workflow and Mapping Systems 12
3.2 Situation Room Operational Architecture 12
3.3 Mapping Systems Register 13
3.4 Situation Room Activation Scaling 14
3.5 Situation Room Outputs by SOP 14
3.6 Data Flow: From Observation to Advisory 14
4. Master data flow across the Early Warning and Monitoring system 16
5. SOP DATA M&A — Data Collection and Analysis 17
5.1 Relevant Annexes 17
5.2 Process Flow Diagram 17
Phase 1: Preparedness 17
Hazard Threshold Escalation Matrix 18
Phase 2: Activation 18
Phase 3: Data Verification 19
Phase 4: Risk Management 20
Phase 5: Data Analysis and Reporting 20
Phase 6: Review and Recovery 20
6. SOP— Monitoring of Multi-Hazard Developments 21
6.1 Relevant Annexes 21

6.2 Four Monitoring Domains 21
6.3 Process Flow Diagram 22
Phase 1: Preparedness 22
Phase 2: Activation 22
Phase 3: Monitoring and Reporting 23
Phase 3b: Anticipatory Risk Briefing and Verification with Member States 23
Phase 4: Synthesis and Interpretation 23
Phase 5: Risk Management and Escalation 24
Phase 6: Review and Recovery 24
7. SOP Watch, Situation Reports and Advisory Issuing 25
7.1 Relevant Annexes 25
7.2 Process Flow Diagrams 25
7.2.1 Watch Process Flow 25
7.2.2 Situation Report Issuing Process Flow 26
7.2.3 Advisory Issuing and Dissemination Process Flow 27
Step 1: Prepare Advisory Draft 28
Step 2: Review and Approval 28
Step 3: Disseminate Advisory 29
Step 4: Confirm Receipt and Escalate Non-Response 29
Step 5: Archive Records 29
7.3 Advisory Escalation Logic 29
7.3.1 Issuing Authority Model 29
7.3.2 Which Product Issued When 30
7.4 Governing Principles 31
7.5 Steps 1–4: Taxonomy, Pathways, Indicators, Thresholds 31
7.6 Steps 5–8: Tiers, Calibration, Governance, Maintenance 31
7.7 Practical Examples 32
8. SOP Multi-Hazard Anticipatory Action Coordination 32
8.1 Relevant Annexes 32
8.2 Process Flow Diagram 33
Step 1: Pre-Disaster Preparedness 33
Step 2: Anticipatory Coordination with NDMAs 33
8.2.1 Coordination Meeting Protocol 33
Step 3: Post-Disaster Review 34
Step 4: Risk Management and Continuous Improvement 34
8.3 After-Hours Protocol 34
SOP ANCILLARIES **Error! Bookmark not defined.**
Annex Master Index **Error! Bookmark not defined.**
Part A: Annex Governance Framework **Error! Bookmark not defined.**
SOP DATA M&A Annexes (Data Collection and Analysis) **Error! Bookmark not defined.**
SOP MONITORING MHD Annexes (Monitoring) **Error! Bookmark not defined.**
SOP WATCH - SRA Annexes (Advisory Issuing) **Error! Bookmark not defined.**

SOP MULTI HAC Annexes (Anticipatory Action) **Error! Bookmark not defined.**
Situation Room Operations Annexes **Error! Bookmark not defined.**
Reference Number Convention **Error! Bookmark not defined.**
Glossary and Abbreviations **Error! Bookmark not defined.**
Document Control **Error! Bookmark not defined.**
Version History **Error! Bookmark not defined.**
Review Schedule **Error! Bookmark not defined.**
References **Error! Bookmark not defined.**

1. Introduction

The Southern African Development Community (SADC) faces diverse and complex multi-hazard risks that include hydrometeorological, geological, biological, and technological hazards. Regional resilience is strengthened when MHEWS technical outputs are operationalised through SADC Humanitarian and Emergency Operations Centre (SHOC) Standard Operating Procedures (SOPs), establishing clear decision triggers, escalation pathways, and coordination protocols while preserving SHOC's autonomous mandate.

This Standard Operating Procedure (SOP) framework outlines the steps, sub-steps, and responsibilities for receiving and consolidating risk information from Member States and recognised regional sources, synthesising hazard intelligence, assessing operational implications, and activating regional coordination processes, as formalised through the SADC MHEWS Memorandum of Understanding (MoU) and associated data-sharing instruments covering both hydrometeorological and non-hydrometeorological hazards.

This SOP is structured around four interconnected procedures aligned to the disaster management cycle. Each SOP is organised by phase (for cyclical monitoring processes in SOPs 2.1 and 2.2) or by step (for sequential action processes in SOPs 2.3 and 2.4), and includes responsible officials, time thresholds, and cross-SOP touchpoints. All SOPs operate through and are supported by the SHOC Situation Room, which serves as the physical and technological hub for multi-hazard monitoring, geospatial analysis, and decision support (see Section 3).

For the purposes of this SOP, 'multi-hazard data' includes hydrometeorological, geological, biological, environmental, technological, and other non-hydrometeorological datasets relevant to disaster risk monitoring and early warning. The procedures outlined below apply equally to slow-onset and rapid-onset hazards, including climate-driven events, public health emergencies, geological hazards, and technological incidents.

1.1 Mandate Clarification

> ⚠ **CRITICAL:** SHOC does not generate forecasts, alerts, or warnings. Forecasting remains the mandate of National Meteorological and Hydrological Services and other authorised specialised institutions. SHOC's role is to consolidate, synthesise, and contextualise in order to produce regional coordination advisories that support anticipatory action and informed decision-making by Member States.

As established in the SHOC Intergovernmental Memorandum of Agreement, SHOC operates as an autonomous regional coordination entity under the SADC Organ on Politics, Defence and Security Cooperation. It is not a forecasting authority and does not deploy or command physical response assets. SHOC's mandate is to facilitate and support regional humanitarian coordination by consolidating and analysing officially issued technical information and working primarily through the designated National Disaster Management Authorities of Member States, while engaging other relevant national agencies in alignment with NDMAs.

1.2 SHOC Institutional Identity (per MoA)

The SHOC Intergovernmental Memorandum of Agreement establishes the following institutional parameters relevant to this SOP:

Parameter	Description
Institutional Status	SHOC is established as an autonomous coordination entity, not a forecasting agency or primary response operator.
Mandate Scope	To coordinate humanitarian and emergency support across SADC Member States primarily through their designated National Disaster Management Authorities (NDMAs) as the entry point, with direct engagement of other relevant national agencies in alignment with NDMAs.
Data Role	SHOC synthesises and interprets data and advisories from NMHS, Regional Specialised Centres, and thematic partners. It does not manage primary ground sensors, satellite feeds, or specialised modelling platforms.
Resource Coordination	SHOC coordinates and advises on the movement of resources. Deployment of physical assets remains the sovereign responsibility of Member States.

1.3 SHOC Staff Designations (per MoA)

Designation	Abbreviation	SOP Role

Executive Director	ED	Strategic focal point for regional political engagement; final approval authority for Red Advisories and full SHOC activation
Head of Programmes	Head Prog.	Second in charge and delegate of the SHOC Executive Director
Head of Operations	Head Ops	Operations decision maker; operational clearance for advisories; Orange Watch approval; Anticipatory Action coordination
Head of Early Warning and Monitoring	Head EW&M	Technical lead for all EW&M SOPs; Yellow Advisory approval; data collection oversight
Data and Information Manager	DIM	Hazard type/region identification; metadata standards; dashboard development
ICT and Security Officer	ICT Officer	Metadata standards; system security; data encryption; platform integration
Field Data Team Lead	FDTL	Field team mobilisation; training; primary data collection protocols
Early Warning and Monitoring Officer	EW&M Officer	Data source identification; verification; routine monitoring cycles
Early Warning Assistants	EW Assistants	Automated and manual data collection; field-based inputs
Liaison Officer	LO	Coordination with Regional/National Focal Points for access and permissions
Information Management and Reporting Officer	IMRO	Cloud repository management; data dissemination; SitRep inputs

1.4 Legal and Institutional Basis

Instrument	Year	Relevance
SADC Protocol on Politics, Defence and Security Cooperation	2001	Governance framework for regional security cooperation
SADC DRM Strategy and Action Plan	(2022–2030)	Strategic direction for DRR activities
SADC Regional Multi-Hazard Early Warning System MoU	2025	Data-sharing agreements and MHEWS coordination
SHOC Establishment Charter and Terms of Reference	2025	Mandate, governance, staff designations, operational authority
SHOC Intergovernmental Memorandum of Agreement (MoA)	2025	Autonomous coordination entity; NDMA relationship
National Disaster Management Acts of SADC Member States	Various	National legal basis for NDMA authority

1.5 Guiding Principles

Principle	Description
People-Centred	SHOC Early Warning and Monitoring processes shall prioritise protection of lives and livelihoods.
Multi-Hazard	Monitoring and coordination mechanisms shall address hydrometeorological, geological, biological, and technological hazards.

End-to-End Approach	Risk monitoring, analysis, coordination, and activation processes shall ensure continuity from risk identification to operational response support.
Interoperability	Data standards and information systems shall remain compatible with national, regional, and recognised international platforms.
Anticipatory Action	Coordination advisories may enable timely preparedness and pre-emptive action by Member States.
NDMA-Primary Engagement	All SHOC engagement with Member States is conducted primarily through NDMAs as the designated entry point, while other relevant national agencies and recognised organisations may be directly engaged in alignment with NDMAs.

GOVERNANCE NOTE: SHOC engagement with Member States is conducted primarily through National Disaster Management Authorities (NDMAs) as the designated entry point. Other relevant national agencies and recognised organisations may be directly engaged in alignment with NDMAs. This ensures alignment with national coordination structures and preserves national sovereignty.

1.6 Reference Standards and Technical Frameworks

Standard	Edition/Year	Application
IFRC Surge Operational Toolbox	2024	Information management methodologies for data collection, verification, and quality assurance processes
UNDAC Handbook	8th Ed. (2024)	Information management, emergency operations centre practices, and reporting standards
WMO Common Alerting Protocol (CAP)	1.2	Informative reference for structured advisory messaging, subject to SHOC mandate alignment (not operationally adopted)
SADC MHEWS Framework	2025	Regional multi-hazard monitoring parameters and escalation thresholds
IFRC Forecast-based Financing Manual	2023	Anticipatory action trigger design and early action planning
OCHA Anticipatory Action Framework	Current	Anticipatory coordination and pre-positioning approaches
ISO 22301:2019	2019	Business continuity management principles
ISO 27001:2022	2022	Information security management and data protection practices
UNHRD Partner Handbook	Current	Resource mobilisation and pre-positioning procedures

1.7 Information Sourcing Hierarchy

In accordance with its coordination mandate, SHOC relies on officially issued information from recognised national, regional, and continental technical authorities. The order of reference for information consolidation under this SOP is as follows:

⚠ **CRITICAL:** The following hierarchy governs all information sourcing under this SOP.

Primary Technical Sources

National Meteorological and Hydrological Services of Member States; Regional Specialised Centres, including Regional Specialised Meteorological Centres (RSMCs), Regional Health Surveillance Hubs, Regional Pest and Transboundary Plant Health Control Bodies, and Seismic Monitoring Centres; SADC Climate Services Centre.

Continental and Regional Economic Community MHEWS Mechanisms

Africa Multi-Hazard Early Warning and Early Action Programme (AMHEWAS); Common Market for Eastern and Southern Africa (COMESA); East African Community (EAC); Intergovernmental Authority on Development (IGAD). Continental and Regional Economic Community MHEWS Mechanisms as information sources are subject to ongoing technical assessment.

Secondary and Supporting Sources

United Nations specialised agencies; Recognised global monitoring platforms, including the Global Disaster Alert and Coordination System (GDACS), Automatic Disaster Analysis and Mapping (ADAM), Pacific Disaster Center (PDC) DisasterAWARE, and MyDewetra.

Note: CAP and AMHEWAS Alignment

The Common Alerting Protocol (CAP) 1.2 is referenced as an informative standard for structured advisory messaging. CAP is not operationally adopted by SHOC and would only become an operational requirement if SHOC formally standardises its coordination advisories using CAP format or integrates with systems that rely on CAP-structured feeds. Advisory templates in this SOP are CAP-aligned as a reference but may be adapted to SHOC's coordination mandate.

The Africa Multi-Hazard Early Warning and Early Action Programme (AMHEWAS) is referenced as a continental alignment standard. SHOC's monitoring parameters are aligned with AMHEWAS standards but AMHEWAS does not serve as a primary data source under this SOP.

1.8 Information Conflict Resolution Protocol

When information from multiple sources within the hierarchy is contradictory, the following resolution procedure applies:

Step	Action	Detail
1	Primary source prevails	Where NMHS or RSMC data conflicts with secondary sources, the primary source data is used for decision-making.
2	Document discrepancy	All discrepancies are recorded in the Data Verification Log (Annex 2.1-C) with source details, timestamps, and nature of conflict.
3	Technical consultation	Head EW&M convenes technical consultation with relevant NMHS or Regional Specialised Centre to resolve material discrepancies.
4	Resolution record	The resolution and its justification are recorded in the Data Verification Log and communicated to all EW&M staff.

2. SHOC Operational Framework for Early Warning and Monitoring

The SHOC Operational Framework for Early Warning and Monitoring is structured across three sequential phases that govern the monitoring and observation of hazard risk information, analytical assessment of operational implications, and the activation of regional coordination processes. Rather than numbered staging systems, SHOC uses descriptive operational phases aligned with UNDAC Emergency Operations Centre practice to provide clarity across diverse national capacities.

2.1 Operational Phase Definitions

The following operational phases describe SHOC's escalating posture. Phase transitions are determined by the hazard threshold levels defined in the Hazard Threshold Escalation Matrix (Annex 2.2-B) and authorised through the Decision Authority Matrix (Section 2.3).

Operational Phase	EW&M Product	Description
Routine Monitoring	Watch	Standard business-hours monitoring of hazard data feeds. Duty EW&M Officer maintains awareness through daily dashboard updates and weekly GIS refresh. Normal working hours with no shift handover requirements.
Heightened Monitoring	Situation Report	Extended monitoring activated when hazard signals elevate. EW&M Officer plus EW Assistant on active duty with DIM on alert. Dashboard refresh every 4 hours with hazard-specific GIS layers activated. Extended hours and on-call arrangements initiated.
Active Response	Advisory	Extended hours with continuous monitoring when risk thresholds approach activation tiers. Full EW&M team with GIS analyst on standby and ICT support. Hourly dashboard refresh with all display stations active. 12-hour shifts initiated with formal handover per Annex 2.2-C.
Full Activation	Anticipatory Action Coordination	24/7 operations with all Situation Room displays active. Full team in rotation with GIS analyst embedded and surge support. Real-time dashboard feeds across all four display walls. 8-hour rotating shifts with formal handover per UNDAC Ch. 11 EOC protocols.

2.2 SHOC Operational Phase Framework

Operational Phase	Scope	EW&M Product	Typical Trigger	Function
Phase I — Monitoring and Observation	Receive and track incoming information from Member States and recognised regional sources	Watch	Elevated hazard signal (Routine)	Awareness
Phase II — Analytical Assessment	Synthesise hazard intelligence and assess potential regional implications	Situation Report	High likelihood / multi-country risk	Assessment
Phase III — Coordination Activation	Translate assessed risk into structured regional coordination action	Advisory	Imminent impact / threshold met	Activation

2.3 Decision Authority Matrix

EW&M Product	Frequency / Hazard Scope	Trigger	Authority (From)	Authority (To)
Watch	Floods, Tropical Cyclones, Drought, Epidemic, Earthquake, Locust Invasion, Food Insecurity	Recommendation section of Watch recommends that a Situation Report be developed	EW&M Officer	Head EW&M
Situation Report	Per hazard type as above; multi-country risk identified	Recommendation section of Situation Report recommends that an	Head EW&M	Head of Ops

		Advisory should be developed		
Advisory	Imminent or confirmed impact; threshold breach confirmed	Threshold breach confirmed through Situation Room composite risk analysis; NDMA coordination required;	Head EW&M	Head of Ops
Coordination Meeting	Multi-country coordination required; anticipatory action triggers approaching	Advisory issued and NDMA coordination meeting convened per SOP MULTI HAC Step 2	Head EW&M	ED

2.4 Cross-SOP Integration

This section maps integration points between SOP for the Early Warning and Monitoring and other SHOC SOPs, as well as internal handoff points in this SOP.

2.4.1 External Integration (Other SOPs)

SOP Name	Integration Points with sections of SHOC SOP on Early Warning and Monitoring
Finance & Administration	Budget release for anticipatory action; expenditure tracking against Decision Authority Matrix
ICT & Telecommunications	Advisory dissemination systems; Situation Room platform integration; communication channels
Emergency Response Teams	ERT activation triggered by Advisory products; operational handoff at coordination activation phase
Human Resources	ERT support activation; Situation Room shift scheduling for 24/7 monitoring during Full Activation
Information Management	Data standards; IMS integration; SHOC Early Warning Products; Situation Room dashboard feeds
Security & Access	Security situation monitoring; access restrictions based on hazard alerts
Logistics & Supply Chain	Pre-positioning triggers from SOP MULTI HAC; resource mobilisation; stock verification. Note: Physical deployment and supply chain operations are governed by SOP Volume 8, not this Volume.

2.4.2 Internal Integration (SOPs 2.1–2.4 Handoff Points)

From SOP	To SOP	Handoff Content	Trigger Mechanism
SOP DATA M&A	SOP MONITORING MHD	Verified data packages with spatial context submitted to Situation Room	SOP DATA M&A Phase 5 outputs feed SOP MONITORING MHD Phase 2 activation
SOP MONITORING MHD	SOP WATCH - SRA	Threshold breach identified; composite risk scores flagged	SOP MONITORING MHD Phase 5 escalation triggers SOP WATCH - SRA Step 1
SOP WATCH - SRA	SOP MULTI HAC	Advisory issued; NDMA coordination required	SOP WATCH - SRA Step 3 dissemination triggers SOP MULTI HAC Step 2
SOP MULTI HAC	SOP DATA M&A	After-Action Review lessons feed back into data collection improvements	SOP MULTI HAC Step 5 updates flow back to SOP DATA M&A Phase 1

3. SHOC Situation Room Operations and Mapping Systems

The SHOC Situation Room is the operational nerve centre for all Early Warning and Monitoring activities under this SOP . It provides the physical and technological environment in which data is observed and monitored, analysed, visually displayed, and synthesised into actionable intelligence for the appropriate SHOC Early Warning Product to be developed and anticipatory action coordination with Member States.

> **GOVERNANCE NOTE:** The Situation Room supports SHOC's synthesis and coordination mandate. SHOC does not manage primary ground sensors, satellite infrastructure, or specialised modelling platforms. The Situation Room integrates analysed products from NMHS, Regional Specialised Centres, and thematic partners.

3.1 Situation Room Workflow and Mapping Systems

The following diagram illustrates the data flow through the SHOC Situation Room, from ingestion of external feeds through analysis to decision support. Worldwide, key Disaster Risk Management (DRM) Situation Rooms include the African Union Commission (AUC), Southern African Development Community (SADC), East African Community (EAC), Economic Community of West African States (ECOWAS), Intergovernmental Authority on Development (IGAD), UN OCHA, ASEAN, the United States' FEMA and USAID, the European Union Civil Protection Mechanism (EUCPM), Asia Pacific Economic Cooperation (APEC), the Caribbean Disaster Emergency Management Agency (CDEMA), and the Mozambique INGD. These rooms coordinate disaster preparedness, response, and recovery across their regions. The SHOC Situation Room operates within a network of regional and national rooms, including AMHEWAS and REC-level Situation Rooms like COMESA, EAC, IGAD, and NDMA National Situation Rooms. External data exchange is structured into one-way feeds from global platforms (e.g., GDACS, ADAM) and reciprocal information sharing with NDMA and REC Situation Rooms for verification and risk interpretation, enabling coordinated regional response.

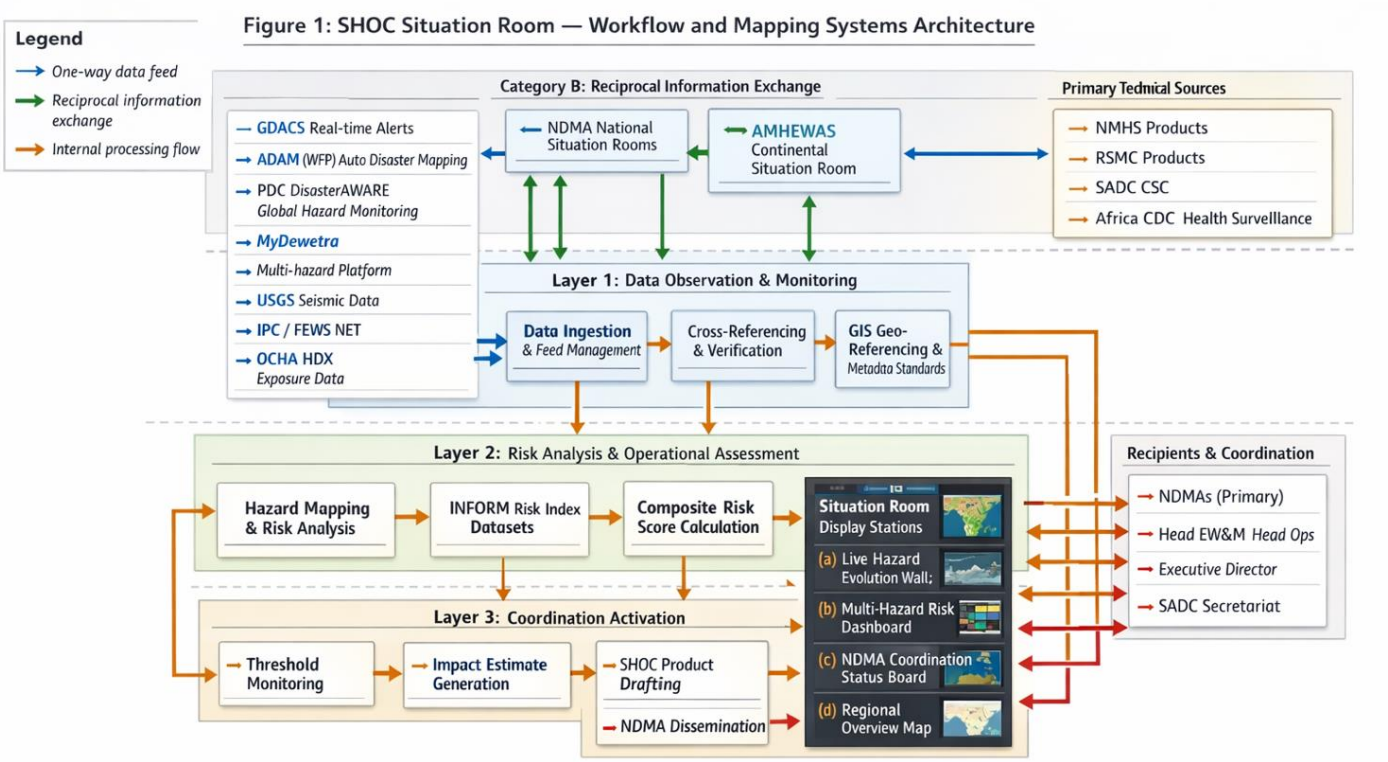

Figure 1: SHOC Situation Room — Workflow and Mapping Systems Architecture

3.2 Situation Room Operational Architecture

The Situation Room operates across three functional layers, each supporting the SOP processes:

Layer	Function	Systems and Tools	SOP Linkage
1. Data Observation and Monitoring	Receive and ingest multi-hazard data feeds from external sources	GDACS real-time alerts; ADAM (WFP); PDC DisasterAWARE; MyDewetra multi-hazard platform; AMHEWAS alignment feeds; NMHS and RSMC products (RSMC La Réunion, RSMC Pretoria); SADC CSC seasonal products; Africa CDC health surveillance; USGS seismic data; IPC/FEWS NET food security; OCHA HDX exposure data; NDMA field reports	SOP DATA M&A: All data collected and verified through this layer (Annexes 2.1-A/B/C/D)
2. Risk Analysis and Operational Assessment	Process, overlay, and analyse spatial data to produce multi-hazard risk composites; render on Situation Room displays and dashboards	**ArcGIS/QGIS** for hazard mapping, exposure mapping, and vulnerability/coping capacity overlays. **SADC CSC GIS Portal** for regional climate layers. **INFORM Risk Index** spatial datasets. **Power BI/Tableau** dashboards with spatial layers. Display stations: (a) Live Hazard Evolution Wall; (b) Multi-Hazard Risk Dashboard; (c) NDMA Coordination Status Board; (d) Regional Overview Map	SOP MONITORING MHD: Four monitoring domains spatially analysed. SOP WATCH - SRA: Advisory drafting draws on displays; GIS-derived maps attached (Annexes 2.2-A/C)
3. Coordination Activation	Translate spatial intelligence into advisory content, threshold breach alerts, and anticipatory action triggers	Threshold monitoring against Annex 2.2-B Escalation Matrix. Automated alerts when composite risk scores approach activation tiers. Impact estimate generation for advisory content. Spatial attachments for NDMA dissemination.	SOP MULTI HAC: Anticipatory actions triggered based on spatial analysis; GIS-derived exposure and impact estimates inform coordination with NDMAs (Annexes 2.4-A/B)

3.3 Mapping Systems Register

The following mapping and geospatial systems are maintained by the Situation Room to develop Early Warning and Monitoring products:

System	Type	Function	Data Refresh	Responsible
ArcGIS / QGIS	GIS Analysis Platform	Multi-hazard risk mapping, exposure analysis, vulnerability overlays, impact estimation, spatial report generation	Per analysis cycle; continuous during Active Response / Full Activation	DIM, Data Analysts
Power BI / Tableau	Dashboard & Visualisation	Situation Room wall displays; spatial layer integration; threshold status visualisation; operational phase tracking	Real-time feeds; hourly refresh minimum during Heightened+	IMRO, ICT Officer
GDACS	Real-Time Global Alerts	Automatic impact estimation for earthquakes, floods, cyclones, droughts; global alert feed ingestion	Real-time (automatic)	EW&M Officer
ADAM (WFP)	Automatic Disaster Mapping	Automated exposure analysis; population in affected areas; infrastructure exposure estimates	Real-time (automatic)	EW&M Officer
PDC DisasterAWARE	Global Hazard Monitoring	Multi-hazard monitoring; early warning visualisation; hazard footprint mapping	Real-time (automatic)	EW&M Officer
MyDewetra	Multi-Hazard Platform	Integrated multi-hazard monitoring, forecast visualisation, and impact assessment for hydrometeorological and geological events	Real-time (automatic)	EW&M Officer

SADC CSC GIS Portal	Regional Climate/Env.	Seasonal forecast spatial layers; vegetation indices; drought monitoring; regional climate products	As published by CSC	DIM
INFORM Risk Index	Exposure & Vulnerability	Pre-positioned risk profiles; vulnerability and coping capacity spatial datasets for all SADC Member States	Annual (pre-loaded); supplemented during activations	Data Analysts

3.4 Situation Room Activation Scaling

The operational tempo and staffing of the Situation Room scales with the operational phase:

Operational Phase	EW&M Product	Staffing	Dashboard Refresh	GIS Analysis Cycle	Shift Protocol
Routine Monitoring	Watch	Duty EW&M Officer; routine DIM support	Daily dashboard; weekly GIS refresh	As required (preparedness)	Normal working hours
Heightened Monitoring	Situation Report	EW&M Officer + EW Assistant; DIM on alert	4-hourly dashboard; hazard GIS activated	Hazard-specific mapping; exposure overlay	Extended hours; on-call
Active Response	Advisory	Full EW&M team; GIS analyst standby; ICT support	Hourly dashboard; all displays active	Continuous spatial analysis; impact estimation per cycle	12-hour shifts; Annex 2.2-C activated
Full Activation	Emergency Coordination	Full team rotation; GIS analyst embedded; surge support	Real-time feeds; all four display walls	Continuous; real-time spatial updates; GIS products attached to all advisories	8-hour rotating shifts; UNDAC Ch. 11

3.5 Situation Room Outputs by SOP

SOP	Situation Room Output	Format	Recipients
SOP DATA M&A	Verified data packages with spatial context; Data Source Registry updates	GIS-annotated datasets; Annex 2.1-A registry entries	Head EW&M; Data Analysts
SOP MONITORING MHD	Multi-hazard risk dashboards; threshold status reports; hazard evolution maps; shift handover logs	Power BI/Tableau dashboards; Annexes 2.2-A/B/C/D products	All EW&M staff; Head Ops; ED
SOP WATCH - SRA	GIS-derived impact maps and exposure estimates attached to advisories; spatial briefing materials	ArcGIS/QGIS map products; spatial annexes to advisory templates (Annex 2.3-A)	NDMAs; Head EW; Head Ops;
SOP MULTI HAC	Threshold breach visualisations; impact-based trigger analysis; resource pre-positioning maps; post-event spatial analysis	Composite risk maps; trigger status displays; Annexes 2.4-A/C products	NDMAs; Head Ops; SADC Secretariat ED (strategic and political engagement only)

3.6 Data Flow: From Observation to Advisory

The following sequence describes the standard data flow through the Situation Room for a typical hazard event, referencing the three-layer operational architecture defined in Section 3.2:

1. Data Observation and Monitoring (Layer 1): Real-time hazard feeds from SHOC approved data observation and Advisory platforms such as MyDewetra, GDACS, ADAM, PDC DisasterAWARE, RSMC products, and NMHS/Regional Specialised Centre products are received and ingested into the SHOC Situation Room GIS (ArcGIS/QGIS) per SOP DATA M&A.

2. Risk Analysis and Operational Assessment (Layer 2): Hazard data is overlaid with exposure layers (population, infrastructure) and vulnerability/coping capacity datasets to produce a multi-hazard risk composite per SOP MONITORING MHD's four monitoring domains. Analysed products are rendered on Situation Room dashboards (Power BI/Tableau with spatial layers), providing real-time situational awareness.

3. Threshold Monitoring: The Multi-Hazard Risk Dashboard tracks composite risk scores against the Hazard Threshold Escalation Matrix (Annex 2.2-B). When thresholds approach activation tiers, alerts are flagged for the duty officer.

4. SHOC Product Drafting (Layer 3): When thresholds are breached, the EW&M Officer drafts the relevant Watch, Situation Report, or Advisory using Situation Room intelligence. GIS-derived impact maps and exposure estimates are attached as spatial annexes per SOP WATCH - SRA.

5. Dissemination with Spatial Products: Advisories are disseminated to NDMAs with GIS-derived spatial attachments showing affected areas, population exposure, infrastructure risk, and recommended anticipatory action zones per SOPs 2.3/2.4.

6. Anticipatory Coordination: During coordination meetings with NDMAs (SOP MULTI HAC, Step 2), Situation Room spatial products provide the evidence base for shared risk interpretation and proportionate action planning.

4. Master data flow across the Early Warning and Monitoring system

The following diagram illustrates the process flows for each SOP and the master data flow across the Early Warning and Monitoring system.

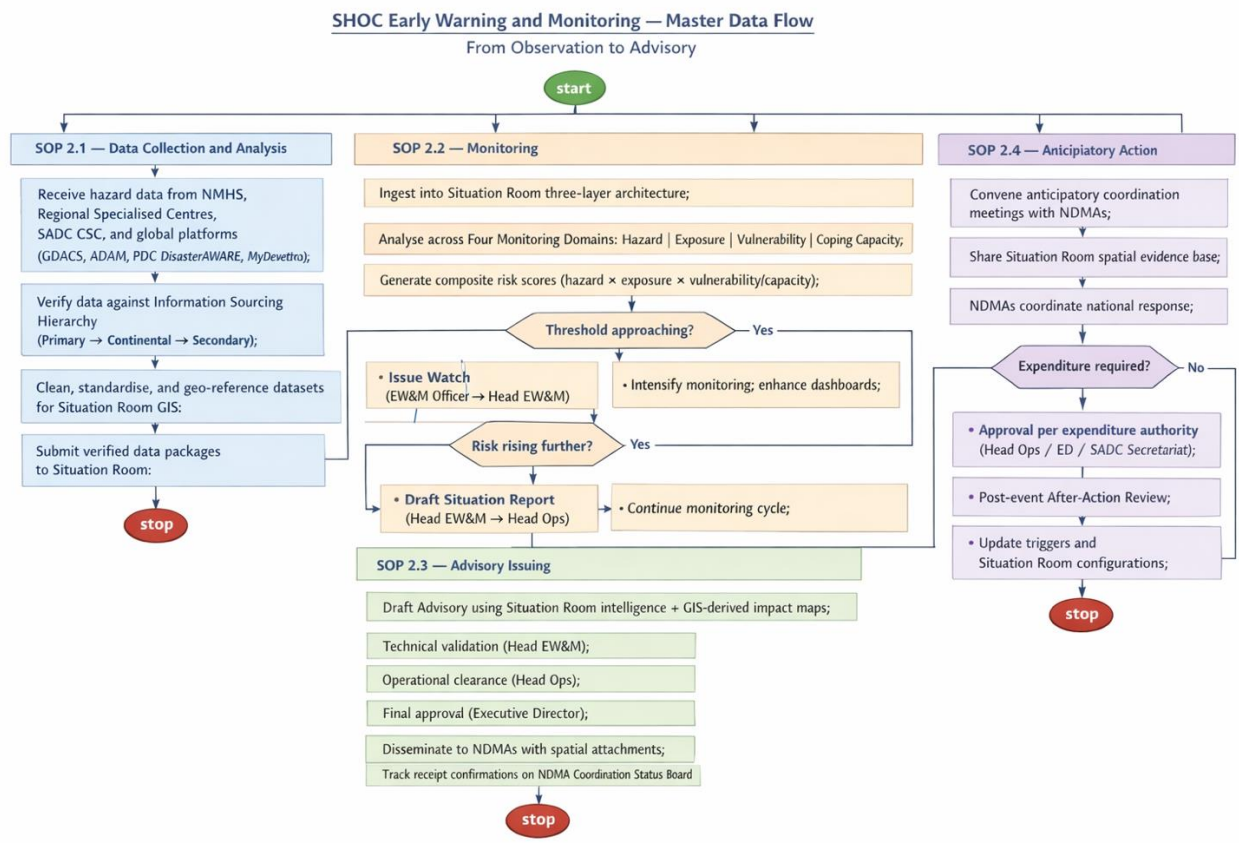

Figure 2: SHOC Early Warning and Monitoring — Master Data Flow

5. SOP DATA M&A — Data Collection and Analysis

This SOP governs the collection, verification, standardisation, and analysis of multi-hazard data from recognised sources. It establishes the procedures for receiving raw data inputs, applying quality assurance protocols, and producing verified data packages for ingestion into the Situation Room (SOP MONITORING MHD). SOP DATA M&A follows a six-phase cyclical structure aligned to the disaster management cycle.

5.1 Relevant Annexes

Annex 2.1-A: Data Source Registry Template [UNDAC 8th Ed. Ch. 9; IFRC IM 2.3]

Annex 2.1-B: Data Collection Checklist — Hydrometeorological and Non-Hydrometeorological [IFRC Surge IM 2.1; SADC MHEWS]

Annex 2.1-C: Data Verification Log — systematic cross-checking (V/D/P status) [UNDAC 8th Ed. Ch. 9.4; IFRC IM 3.2]

Annex 2.1-D: Data Quality Assurance Protocol — Accuracy (≥95%), Completeness (≥90%), Timeliness (≤6h), Consistency (0% variance), Validity (100%) [ISO 27001:2022; IFRC IM 4.1]

5.2 Process Flow Diagram

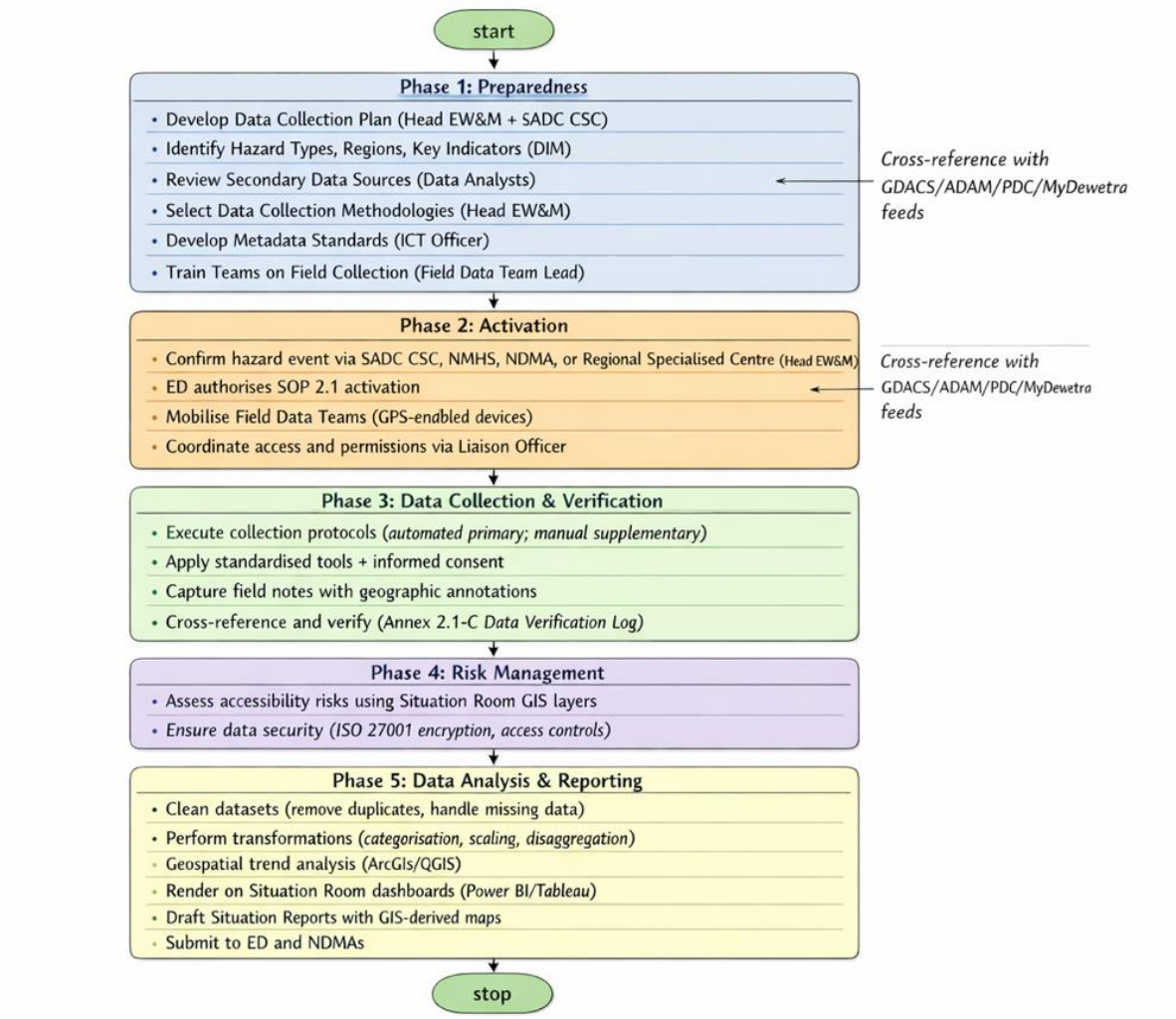

Figure 3: SOP— Data Collection and Analysis

Phase 1: Preparedness

Sub-Step	Description	How To	Responsible	Outcome	Time
1.1	Data Collection Plan Development	Develop plan covering all multi-hazard data types, sources, methodologies, schedules. Include spatial data requirements for Situation Room GIS.	Head EW&M + SADC CSC	Approved plan	Prep.
1.2	Identify Hazard Types, Regions, Key Indicators	Map hazard types to regions; define monitoring indicators; establish GIS layer requirements per hazard-region combination.	Data & Information Manager	Hazard–region matrix	≤3 days
1.3	Review Secondary Data Sources	Catalogue secondary data including historical records, partner databases, INFORM Risk Index spatial datasets, OCHA HDX exposure layers.	Data Analysts	Data inventory	Ongoing
1.4	Select Data Collection Methodologies	Define survey, interview, focus group, automated collection methodologies.	Head EW&M	Methodology framework	Prep.
1.5	Develop Metadata Standards	Establish metadata including geographic reference standards ensuring GIS interoperability across national/regional systems.	ICT Officer	Metadata standards	Prep.
1.6	Train Teams on Field Collection	Train on standardised protocols, informed consent, data security, GPS-referenced field reporting.	Field Data Team Lead	Trained teams	Quarterly

Hazard Threshold Escalation Matrix

The following matrix defines threshold levels for SHOC Early Warning Products by hazard type. Full thresholds are maintained in Annex 2.2-B

Hazard Type	Watch	Situation Report	Advisory	Escalation Pathway
Tropical Cyclone	72h landfall; <Cat 1	48h landfall; Cat 1–2	24h landfall; Cat 3+	Routine → Heightened → Active → Full
Riverine Flood	>Alert level	>Warning level	>Danger level	Routine → Heightened → Active → Full
Drought	SPI -1.0 to -1.5	SPI -1.5 to -2.0	SPI < -2.0	Routine → Heightened → Active → Full
Earthquake	—	Mw 5.0–5.9	Mw ≥6.0	Heightened → Active → Full
Epidemic	Cases >baseline	>Epidemic threshold	Outbreak declared	Routine → Heightened → Active → Full
Food Insecurity	IPC Phase 2	IPC Phase 3	IPC Phase 4+	Routine → Heightened → Active → Full

Phase 2: Activation

Sub-Step	Description	How To	Responsible	Outcome	Time

2.1	Confirm Hazard Event	Confirm via SADC CSC, NMHS, NDMA, or Regional Specialised Centre. Upon activation, the Situation Room transitions to 24-hour operations as the coordination centre for all Early Warning and Monitoring activities. Cross-reference with GDACS/ADAM/PDC DisasterAWARE/MyDewetra real-time feeds on Situation Room displays.	Head EW&M	Event confirmed	ASAP (target: 2h)
2.2	SOP DATA M&A	Seek formal authorisation from Head of Operations.	Head EW&M → Head Ops	SOP DATA M&A activated	ASAP (target: 1h)
2.3	Mobilise Field Data Team	Deploy field teams; ensure GPS-enabled data collection devices issued.	Field Data Team Lead	Teams deployed	ASAP (target: 6h)
2.4	Coordinate Access/Permissions	Liaise with Regional/National Focal Points for access, security, logistics.	Liaison Officer	Access confirmed	ASAP (target: 4h)

Phase 3: Data Verification

Sub-Step	Description	How To	Responsible	Outcome	Time
3.1	Execute Collection Protocols	Execute survey/interview/focus group protocols. Verification of data acquired under this Phase is confined to Early Warning and Monitoring purposes for the production of Situation Room products. Data verification follows the protocols established in Annex 2.1-C, with cross-referencing against primary technical sources defined in Section 1.7. Automated systems primary; manual/field supplementary. Ensure all data GPS-referenced for Situation Room GIS ingestion.	EW&M Officer, EW Assistants	Raw dataset	Daily
3.2	Standardised Tools + Consent	Apply standardised instruments; ensure informed consent; ethical guidelines.	Field Data Team	Clean data	Per cycle
3.3	Capture Field Notes	Document anomalies, issues, challenges with geographic annotations.	Field Data Team	Annotated dataset	Per cycle

Phase 4: Risk Management

Sub-Step	Description	How To	Responsible	Outcome	Time
4.1	Assess Accessibility Risks	Evaluate risks using Situation Room GIS layers (infrastructure damage, conflict zones, environmental hazards).	Head EW&M	Risk assessment	Before deployment
4.2	Ensure Data Security	Encryption, access controls, secure transmission per ISO 27001:2022.	ICT Officer	Secured environment	Continuous

Phase 5: Data Analysis and Reporting

Sub-Step	Description	How To	Responsible	Outcome	Time
5.1	Clean Datasets	Remove duplicates, handle missing data (R, SPSS, Excel).	Data Analysts	Clean dataset	ASAP (target: 4h)
5.2	Perform Transformations	Categorisation, scaling, disaggregation for cross-hazard comparison.	Data Analysts	Transformed dataset	ASAP (target: 6h)
5.3	Trend Analysis and Impact Assessments	ArcGIS/QGIS for geospatial mapping; R/SPSS for statistical analysis; produce multi-hazard risk composite layers for Situation Room displays.	Senior Analyst, EW&M Officer	Impact analysis + GIS products	ASAP (target: 6h)
5.4	Create Dashboards	Render analytical outputs on Situation Room Power BI/Tableau dashboards with spatial layers.	DIM, IMRO	Updated dashboards	End of day
5.5	Draft SitReps	Prepare Situation Reports with GIS-derived maps and impact summaries.	IMRO	SitRep + spatial annexes	End of day
5.6	Submit to ED and NDMAs	Submit analytical products with spatial context to ED and to NDMAs via NDMA channels.	Head EW&M	Reports disseminated	End of day

Phase 6: Review and Recovery

Sub-Step	Description	How To	Responsible	Outcome	Time
6.1	After-Action Review	Structured AAR with NDMAs; include spatial accuracy assessment.	Head EW&M	AAR report	≤2 days
6.2	Update SOP DATA M&A	Integrate lessons including GIS/mapping system improvements.	Head EW&M	Updated SOP	≤1 week
6.3	Upgrade Systems	Assess and upgrade data management, GIS platforms, dashboard configurations.	ICT Officer, DIM	Upgrade plan	≤2 weeks

6. SOP— Monitoring of Multi-Hazard Developments

This SOP governs the continuous monitoring, analysis, and synthesis of multi-hazard information within the SHOC Situation Room. It establishes procedures for tracking hazard evolution across four monitoring domains, producing composite risk assessments, and determining when escalation to advisory products is required. SOP MONITORING MHD follows a six-phase cyclical structure with an additional anticipatory briefing phase (Phase 3b).

6.1 Relevant Annexes

Annex 2.2-A: Multi-Hazard Monitoring Dashboard Parameters [AMHEWAS Standards (alignment only)]

Annex 2.2-B: Hazard Threshold Escalation Matrix [SADC MHEWS; IFRC EWEA; WMO Impact-Based Methods]

Annex 2.2-C: Shift Handover Log [UNDAC 8th Ed. Ch. 11 (EOC Operations)]

Annex 2.2-D: Hazard Evolution Assessment Report [WMO Impact-Based Methods]

6.2 Four Monitoring Domains

Domain	Description	Situation Room Display	Key GIS Layers
1. Hazard	Real-time and assessed hazard data	Live Hazard Evolution Wall: cyclone tracks, flood extent, seismic zones, epidemic spread, drought indices	RSMC products; GDACS/ADAM/PDC/MyDewetra feeds; NMHS national products; health surveillance spatial data
2. Exposure	Population, infrastructure, livelihoods in hazard-prone areas	Regional Overview Map: population density, critical infrastructure, service delivery points in assessment footprint	National statistical office data; OCHA HDX; INFORM Risk Index; settlement and infrastructure inventories
3. Vulnerability	Socio-economic, environmental, health, institutional factors	Multi-Hazard Risk Dashboard: vulnerability composite scores, IPC classifications, health capacity indicators	IPC/FEWS NET spatial layers; WHO AFRO (secondary); poverty indices; displacement tracking
4. Coping Capacity	Response capacity, preparedness, early action mechanisms	NDMA Coordination Status Board: readiness scores, resource availability, response team status	NDMA readiness assessments; stockpile locations; IFRC capacity mapping

6.3 Process Flow Diagram

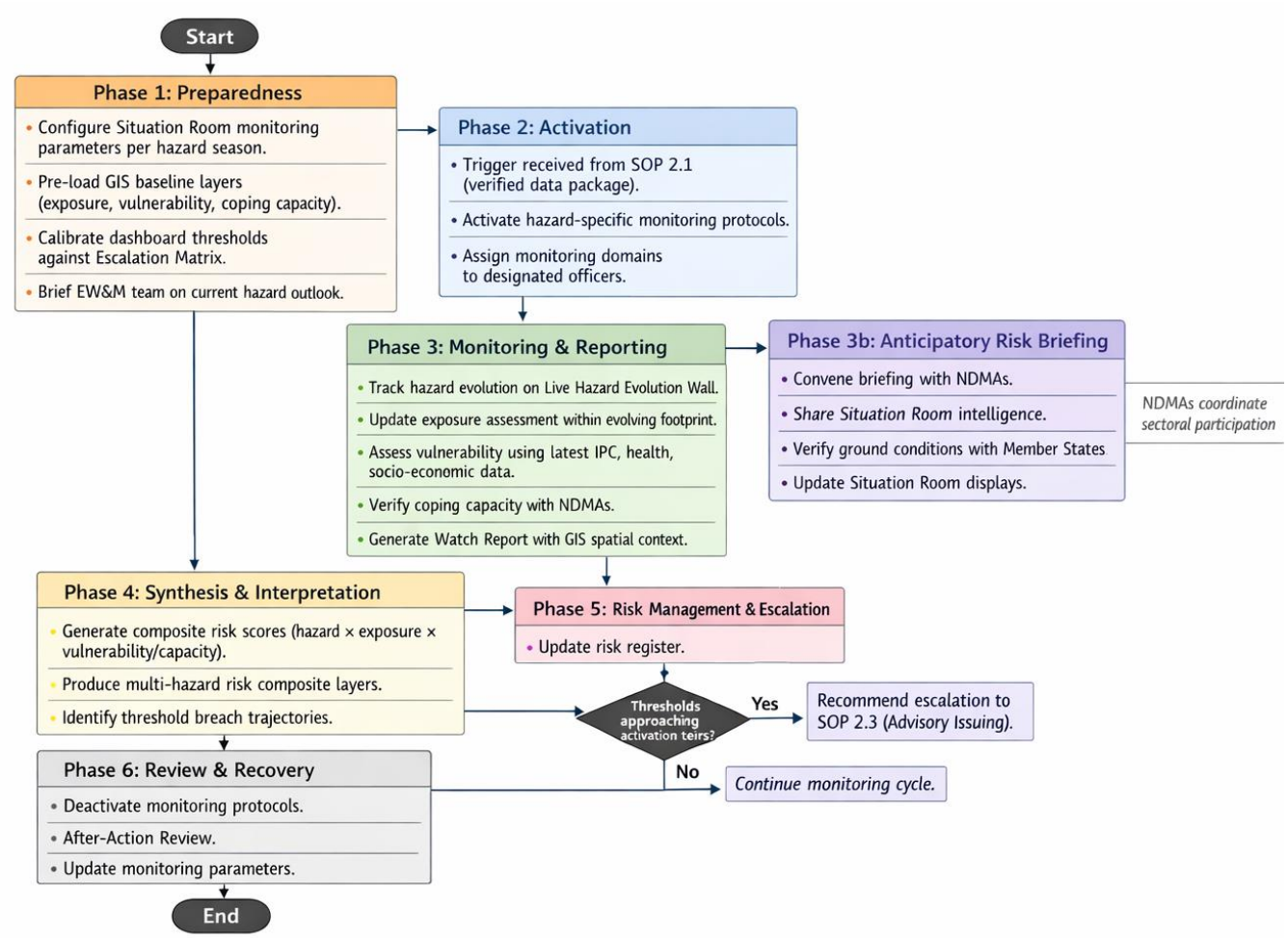

Figure 4: SOP MONITORING MHD — Monitoring of Multi-Hazard Developments

Phase 1: Preparedness

Sub-Step	Description	How To	Responsible	Outcome	Time
1.1	Configure Monitoring Parameters	Set Situation Room monitoring parameters per hazard season using Annex 2.2-A dashboard specifications.	Head EW&M	Configured dashboards	Pre-season
1.2	Pre-load GIS Baselines	Load baseline exposure, vulnerability, and coping capacity layers into Situation Room GIS.	DIM, Data Analysts	GIS baselines active	Pre-season
1.3	Calibrate Thresholds	Review and calibrate dashboard thresholds against Annex 2.2-B Escalation Matrix.	Head EW&M, DIM	Calibrated thresholds	Pre-season
1.4	Brief EW&M Team	Brief all EW&M staff on current hazard outlook, monitoring priorities, and shift arrangements.	Head EW&M	Team briefed	Pre-season

Phase 2: Activation

Sub-Step	Description	How To	Responsible	Outcome	Time
2.1	Receive Trigger from SOP DATA M&A	Verified data package received from SOP DATA M&A indicating elevated hazard signal.	EW&M Officer	Trigger received	Per SOP DATA M&A
2.2	Activate Monitoring Protocols	Activate hazard-specific monitoring protocols; increase dashboard refresh rate.	Head EW&M	Protocols active	ASAP (target: 1h)
2.3	Assign Domain Monitoring	Assign monitoring domains (hazard, exposure, vulnerability, coping capacity) to designated officers.	Head EW&M	Domains assigned	ASAP (target: 1h)

Phase 3: Monitoring and Reporting

Sub-Step	Description	How To	Responsible	Outcome	Time
3.1	Track Hazard Evolution	Monitor real-time hazard feeds on Live Hazard Evolution Wall; update tracks, extents, and intensity metrics.	EW&M Officer	Hazard update log	Continuous
3.2	Update Exposure Assessment	Refresh population and infrastructure exposure within evolving hazard footprint.	Data Analysts	Exposure update	Per cycle
3.3	Assess Vulnerability	Update vulnerability composite scores using latest available data from IPC, health, and socio-economic sources.	Data Analysts	Vulnerability update	Per cycle
3.4	Verify Coping Capacity	Contact NDMAs to verify readiness status, resource availability, and response team positioning.	Liaison Officer	Coping capacity update	Per cycle
3.5	Generate Watch Report	Compile monitoring outputs into Watch report format with GIS-derived spatial context.	EW&M Officer	Watch report	12–24h cycle

Phase 3b: Anticipatory Risk Briefing and Verification with Member States

GOVERNANCE NOTE: SHOC convenes briefings with NDMAs. NDMAs coordinate sectoral participation per national arrangements.

Sub-Step	Description	How To	Responsible	Outcome	Time
3b.1	Convene NDMA Briefing	Present Situation Room intelligence and spatial products to NDMAs for ground-truth verification.	Head EW&M	Briefing completed	As required
3b.2	Verify Ground Conditions	NDMAs provide field-level verification of hazard conditions, exposure status, and local capacity.	Liaison Officer	Verification received	ASAP (target: 4h)
3b.3	Update Situation Room	Incorporate NDMA ground-truth data into Situation Room displays and risk composites.	DIM	Displays updated	ASAP (target: 2h)

Phase 4: Synthesis and Interpretation

Sub-Step	Description	How To	Responsible	Outcome	Time
4.1	Generate Composite Risk Scores	Calculate composite risk scores: hazard × exposure × vulnerability/coping capacity using Situation Room GIS layers.	Data Analysts, DIM	Composite risk scores	Per cycle
4.2	Produce Risk Composite Layers	Generate multi-hazard risk composite spatial layers for Situation Room display.	GIS Analyst	Composite GIS layers	Per cycle
4.3	Identify Threshold Trajectories	Assess whether composite risk scores are approaching activation tiers per Annex 2.2-B.	EW&M Officer	Threshold trajectory assessment	Per cycle

Phase 5: Risk Management and Escalation

Sub-Step	Description	How To	Responsible	Outcome	Time
5.1	Update Risk Register	Record current risk levels, threshold proximity, and recommended actions.	Head EW&M	Updated risk register	Per cycle
5.2	Escalate if Thresholds Approaching	If composite risk scores approach activation tiers, recommend escalation to SOP WATCH - SRA (Advisory Issuing).	Head EW&M → Head Ops	Escalation decision	ASAP

Phase 6: Review and Recovery

Sub-Step	Description	How To	Responsible	Outcome	Time
6.1	Deactivate Monitoring	Return to Routine Monitoring phase when hazard risk subsides.	Head EW&M	Deactivation confirmed	Per assessment
6.2	After-Action Review	Structured AAR reviewing monitoring performance, threshold accuracy, and Situation Room effectiveness.	Head EW&M	AAR report	≤1 week
6.3	Update Parameters	Revise monitoring parameters, thresholds, and GIS layer specifications based on lessons learned.	Head EW&M, DIM	Updated parameters	≤2 weeks

7. SOP Watch, Situation Reports and Advisory Issuing

This SOP governs the preparation, approval, dissemination, and archiving of SHOC Early Warning and Monitoring products: the Watch, Situation Report, and Advisory. Each product corresponds to a distinct stage of risk evolution (see Section 7.3) and follows a sequential workflow triggered by escalation from SOP MONITORING MHD. It establishes the procedures for translating Situation Room intelligence into structured coordination products for dissemination to NDMAs. This SOP should be read in conjunction with the Regional Anticipatory Action Working Group (RAAWG) framework currently under development, which will provide additional guidance on anticipatory action coordination and advisory product standardisation across the SADC region.

Situation Room Integration: All three EW&M products (Watch, Situation Report, and Advisory) draw directly on Situation Room intelligence. GIS-derived impact maps and exposure estimates from the Situation Room's geospatial analysis layer are attached as spatial annexes to all products, providing NDMAs with actionable geographic context for their national coordination.

7.1 Relevant Annexes

Annex 2.3-A: Advisory Message Template (CAP-aligned reference, not adopted) [WMO CAP; OASIS CAP 1.2; IFRC Public Warning]

Annex 2.3-B: Advisory Approval Workflow Form [IFRC EWEA; UNDAC Ch. 12]

Annex 2.3-C: Dissemination Channel Matrix [OCHA IM Guidelines]

Annex 2.3-D: Advisory Receipt Confirmation Log [UNDAC 8th Ed. Ch. 12]

7.2 Process Flow Diagrams

The following process flow diagrams illustrate the sequential workflows for the three SHOC Early Warning and Monitoring products: Watch, Situation Report, and Advisory. Each diagram reflects the escalation logic defined in Section 7.3, with product-specific approval pathways, dissemination procedures, and escalation decision points.

7.2.1 Watch Process Flow

The Watch is the first product in the SHOC early warning chain, issued during the Routine Monitoring operational phase when elevated hazard signals are detected. The Watch process flow encompasses continuous monitoring of hazard evolution through the Situation Room (SOP MONITORING MHD, Phase 3), compilation of monitoring outputs into a structured Watch report on a 12–24 hour cycle, technical validation by the Head of EW&M, dissemination to NDMAs, and an escalation decision point for recommending transition to a Situation Report. The Watch approval pathway requires EW&M Unit validation only (EW&M Officer → Head EW&M). Relevant annexes: Annex 2.2-A (Dashboard Parameters), Annex 2.2-B (Threshold Escalation Matrix), Annex 2.3-C (Dissemination Channel Matrix).

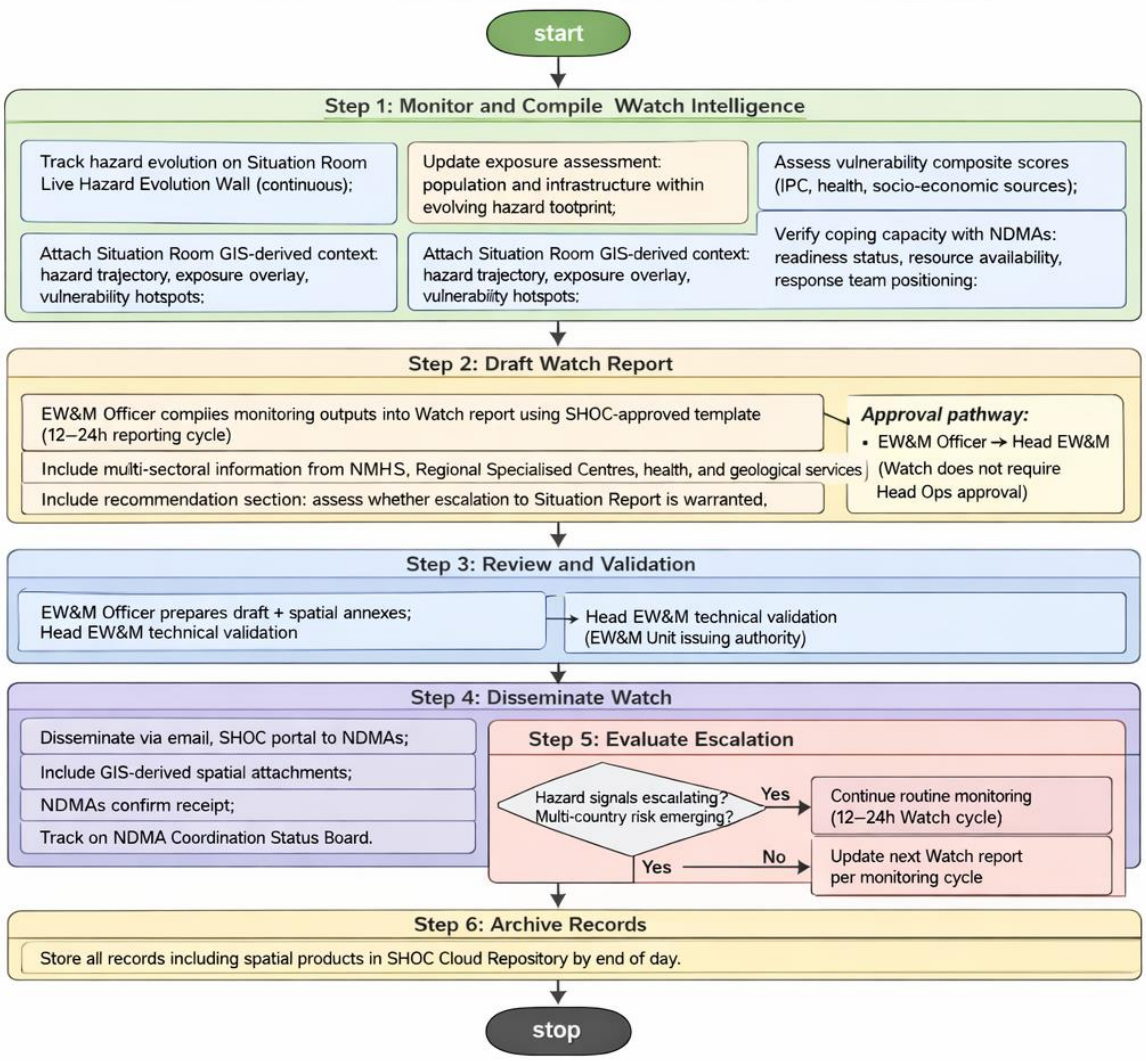

Figure 5: SOP WATCH - SRA — Watch Issuing Process Flow [Source: watch_process_flow.puml]

7.2.2 Situation Report Issuing Process Flow

The Situation Report is the second product in the SHOC early warning chain, issued during the Heightened Monitoring operational phase when risk is rising and multi-country impacts are plausible. The Situation Report process flow is triggered by a Watch recommendation or by SOP MONITORING MHD Phase 5 escalation when composite risk scores approach activation tiers. It encompasses multi-hazard risk composite analysis, GIS-derived impact assessments, drafting by the IMRO with spatial annexes, a mid-level approval pathway (Head EW&M → Head of Operations), dissemination on a 6–24 hour update cycle, a receipt escalation cascade (1h → 2h → 4h), and an escalation decision point for recommending transition to an Advisory. Relevant annexes: Annex 2.2-B (Threshold Escalation Matrix), Annex 2.3-A (Advisory Message Template), Annex 2.3-B (Approval Workflow Form), Annex 2.3-C (Dissemination Channel Matrix), Annex 2.3-D (Receipt Confirmation Log).

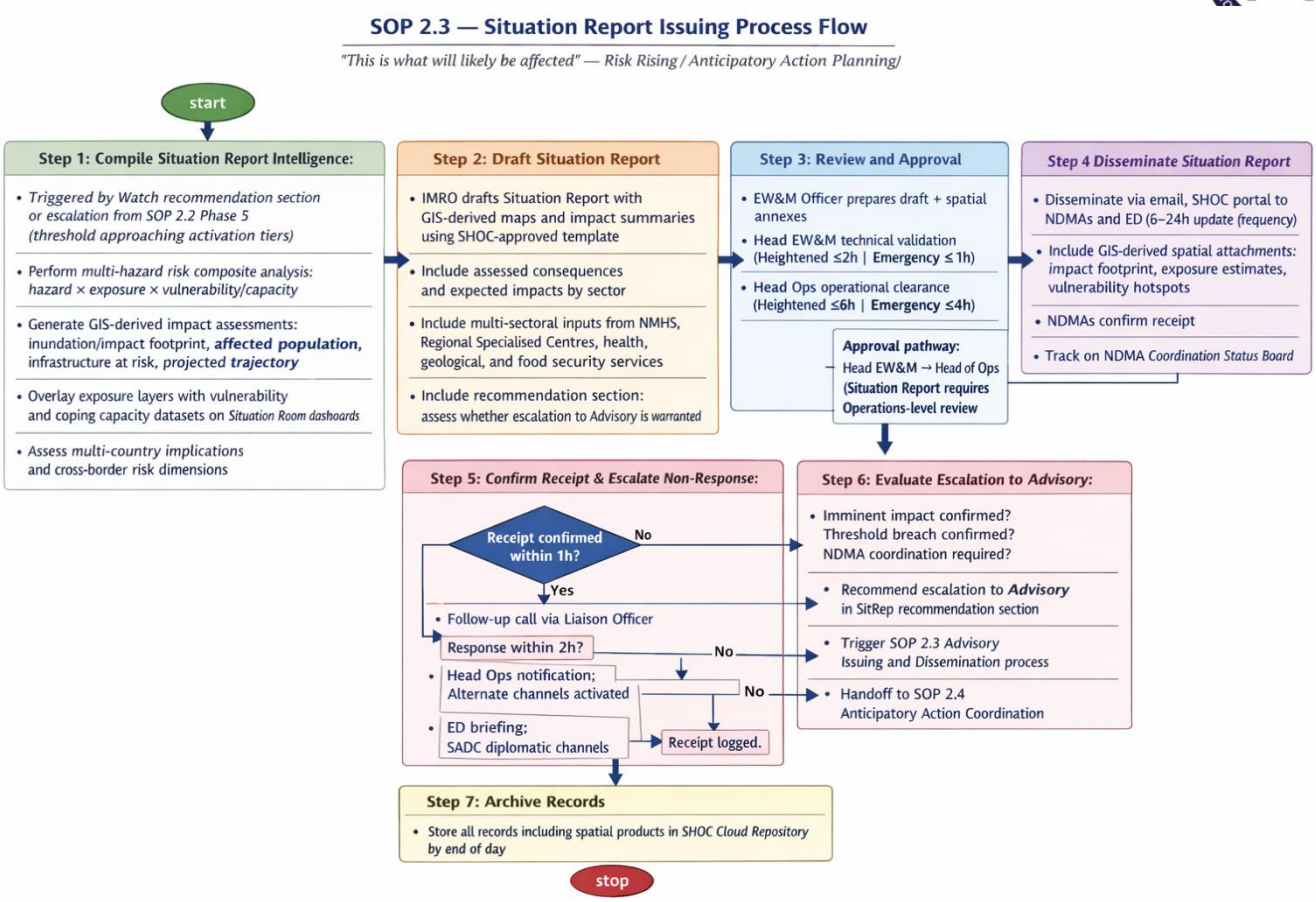

Figure 6: SOP WATCH - SRA — Situation Report Issuing Process Flow

7.2.3 Advisory Issuing and Dissemination Process Flow

The Advisory is the third and highest-level product in the SHOC early warning chain, issued during the Active Response operational phase when imminent impact is confirmed or threshold breaches have been verified. The Advisory process flow is triggered by a Situation Report recommendation or confirmed threshold breach through Situation Room composite risk analysis. It requires the full approval pathway (EW&M Officer → Head EW&M → Head of Operations → Executive Director for Emergency level), and includes the receipt escalation cascade and handoff to SOP MULTI HAC (Anticipatory Action Coordination). Relevant annexes: Annex 2.3-A (Advisory Message Template), Annex 2.3-B (Approval Workflow Form), Annex 2.3-C (Dissemination Channel Matrix), Annex 2.3-D (Receipt Confirmation Log).

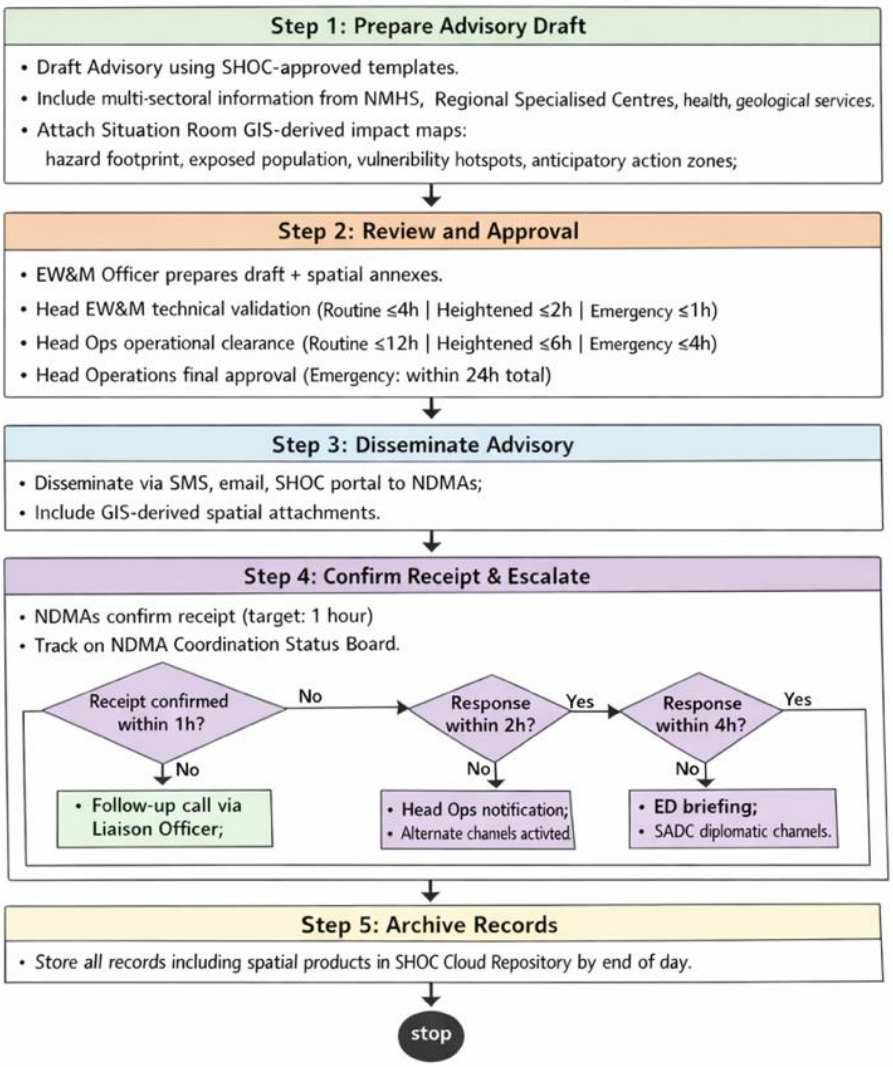

Figure 7: SOP WATCH - SRA — Advisory Issuing and Dissemination

Step 1: Prepare Advisory Draft

Draft using SHOC-approved internal templates. Include multi-sectoral information from NMHS, Regional Specialised Centres, health, and geological services. Attach Situation Room GIS-derived impact maps showing: (a) hazard footprint, (b) exposed population and infrastructure, (c) vulnerability hotspots, (d) recommended anticipatory action zones.

Step 2: Review and Approval

Stage	Action	Responsible	Time by Operational Phase
1	Draft Advisory + spatial annexes prepared	EW&M Officer	—
2	Technical validation (including spatial products)	Head of EW&M	Routine: ≤4h | Heightened: ≤2h | Emergency: ≤1h
3	Operational clearance	Head of Operations	Routine: ≤2h | Heightened: ≤1h | Emergency: ≤30min
4	Final approval	Executive Director	Emergency: within 1.5h total

5	Dissemination authorised	ICT / Comms Officer	—

Step 3: Disseminate Advisory

Disseminate via email, SHOC portal, and social media (as appropriate) to NDMAs. Include spatial attachments (GIS-derived maps).

> **GOVERNANCE NOTE:** Participation of sectoral authorities is coordinated by the respective NDMA.

Step 4: Confirm Receipt and Escalate Non-Response

NDMAs confirm receipt as soon as practicable. Track on Situation Room NDMA Coordination Status Board. Where receipt is not confirmed within 1 hour, the following escalation procedure applies:

Elapsed Time	Status	Responsible	Action
1 hour	No confirmation received	Liaison Officer	Follow-up call to NDMA primary contact via designated communication channel.
2 hours	Still no response after follow-up	Head of Operations	Head Ops notified; alternative communication channels activated (secondary contact, regional focal point).
4 hours	No response via any channel	Executive Director	ED briefed; decision on engagement through SADC Secretariat diplomatic channels.
6 hours	Extended non-response	ED / Head EW&M	Documented in Advisory Receipt Confirmation Log (Annex 2.3-D); flagged for After-Action Review.

Step 5: Archive Records

Store all records including spatial products in SHOC Cloud Repository by end of day.

7.3 Advisory Escalation Logic

The following table defines the escalation logic governing SHOC advisory products. Each product type corresponds to a distinct stage of risk evolution and serves a specific decision-support function within the early warning chain.

Stage	Product	Focus	Decision Use
Early risk	Watch	Hazard probability and trajectory	Readiness and preparedness activation
Risk rising	Situation Report	Assessed consequences and expected impacts	Anticipatory action planning
Event active	Advisory	Confirmed impacts and coordination requirements	Regional coordination activation

Conceptual Distinction: Watch = 'Something may happen.' Situation Report = 'This is what will likely be affected.' Advisory = 'This is what has happened and what we are coordinating.'

7.3.1 Issuing Authority Model

Product	Issuing Authority	Update Frequency
Watch	EW&M Unit validation (Head EW&M)	12–24 hours
Situation Report	EW&M + Operations review (Head Ops)	6–24 hours

Advisory	EW&M + Operations + Leadership (ED)	6–24 hours

All products released through NDMA channels in intergovernmental systems.

7.3.2 Which Product Issued When

Hazard	Watch (issued when…)	Situation Report (issued when…)	Advisory (issued when…)
Tropical Cyclone	A credible TC forms/strengthens and forecast track cone shows **possible** impact to SADC coastal states within **72–120 hrs**; model agreement improving; marine hazards rising.	Landfall/close approach becomes **likely within 72 hrs**, or significant impacts begin (gales, storm surge, heavy rain). Include track, intensity trend, exposed districts, evacuation status, port/airport status.	When protective actions are required: evacuations, port closure guidance, movement restrictions, pre-positioning, ERT standby/activation. Typically **48–72 hrs pre-impact**, and updated as track/intensity shifts.
Flood (flash / pluvial)	Heavy rainfall forecast with high confidence; soils saturated; urban drainage exceedance likely; convective nowcasting indicates rapid onset risk within **0–24 hrs**.	Flooding observed/confirmed (reports, gauges, satellite/radar) with impacts to people/infrastructure. Include inundation footprint, road cut-offs, shelter needs, WASH risks.	When the public/authorities must act: move to higher ground, protect critical facilities, school closure, road closure, evacuation of low-lying settlements. Often **immediate** for flash floods.
Riverine Flood	River levels rising with forecast exceedance of **alert/warning thresholds** within **24–120 hrs**; upstream rainfall and basin routing indicate high probability of overflow.	Thresholds exceeded or overflow underway; impacts to bridges, farms, settlements; cross-border basin implications. Include gauge status, expected crest time, affected wards, safe routes.	When coordinated actions are required: controlled releases, evacuation of floodplains, protection of waterworks/clinics, logistics routing changes, cross-border notifications (basin states).
Earthquake	A "watch" is generally **not used** for tectonic earthquakes (no reliable short-term prediction). Use a **Seismic Information Notice** if heightened regional seismicity or volcanic-related swarm is detected.	Immediately after an event **M≥5** (or lower if shallow/near population) or confirmed damage. Include magnitude, depth, epicentre, shaking intensity map, aftershock probability, initial impact reports.	When guidance is needed: aftershock safety, building inspections, tsunami guidance (if coastal), activation of SAR/INSARAG interfaces, critical infrastructure shutdown protocols.
Epidemic	Signals of abnormal increase (EWARN/IDSR triggers): clustering, positivity rate rise, **R· > 1**, rising severe cases, cross-border importation risk, lab confirmation pending.	Confirmed outbreak or sustained transmission; include epi-curve, attack rate, hotspots (GIS), health system capacity, case fatality trend, response actions underway.	When public health actions must be implemented: IPC measures, vaccination campaign guidance, travel/entry screening decisions (as mandated), risk communication, activation of health ERT/medical surge.
Food Security	Early stress signals: rainfall deficit/vegetation anomalies, market price spikes, livelihood disruption, production shortfalls, IPC projected deterioration in **1–3 months**.	IPC analysis confirms **Crisis/Emergency** (or rapid deterioration) and impacts are measurable: GAM rates, coping strategies, displacement, pipeline gaps.	When programmatic decisions/actions are required: anticipatory action triggers, scale-up of cash/food assistance, nutrition surge, logistics corridors prioritised, cross-border trade facilitation requests.

Methodology for establishing, calibrating, and maintaining anticipatory action triggers. Thresholds are monitored through the Situation Room Multi-Hazard Risk Dashboard.

7.4 Governing Principles

Principle	Description
Impact-based, not hazard-only	Thresholds link to consequences (people/assets/services), not only hazard metrics.
Tiered activation	Multiple thresholds (Watch/Situation Report/Advisory) to avoid all-or-nothing decisions.
Data maturity aware	Where high-resolution data is missing, use proxy indicators and expert judgement.
Regionally agreed and auditable	Every trigger has definition, source, calculation, and review schedule. Regional thresholds are formally endorsed by Member States through the Governing Board; national thresholds may be refined to suit local context.
Spatially referenced	All triggers linked to geographic areas; Situation Room GIS tracks threshold status by location.

7.5 Steps 1–4: Taxonomy, Pathways, Indicators, Thresholds

Step 1 — Hazard Taxonomy: Hydromet (flood, drought, cyclone, heatwave); Geological (earthquake, landslide); Biological (cholera, zoonotic); Technological (spills, dam failure, infrastructure). Output: Hazard Trigger Catalogue.

Step 2 — Impact Pathways: For each hazard, map the causal chain from hazard event through exposure and vulnerability to consequences (lives, livelihoods, infrastructure, services). Output: Impact Pathway Diagrams per hazard type. Per hazard: flood → displacement; heatwave → mortality; outbreak → doubling rate; landslide → isolation. Output: Impact Pathway Map.

Step 3 — Select Indicators (3 Layers): Layer 1 Hazard (rainfall, river level, seismic, case incidence); Layer 2 Exposure (population, infrastructure, settlements); Layer 3 Vulnerability/Capacity (poverty, health capacity, readiness). All layers mapped spatially in Situation Room GIS. Output: Indicator Matrix.

Step 4 — Threshold Types: Absolute thresholds (fixed values, e.g., river level > X metres); Percentile thresholds (relative to historical distribution, e.g., rainfall > 90th percentile); Composite thresholds (multi-indicator triggers combining hazard, exposure, and vulnerability layers); Time-based thresholds (rate of change or sustained exceedance). Output: Threshold Definition Table per hazard type. Absolute (river > X m); Percentile (> 90th); Rate-of-change (doubling < 7 days); Compound (rain + saturation + exposure); Confidence-based (≥60% probability). Output: Threshold Definition Sheet.

7.6 Steps 5–8: Tiers, Calibration, Governance, Maintenance

Step 5 — Tier selection:

Tier	Operational Phase Transition	Actions
Tier 1: Watch Readiness	Routine → Heightened	Monitoring intensified, partners alerted, Situation Room dashboards enhanced.
Tier 2: Alert / Pre-position	Heightened → Active	Pre-position resources, standby rosters, targeted messaging, GIS analyst on standby.
Tier 3: Activate Anticipatory Action	Active → Full	Release funds, coordinate deployment, initiate early actions, full Situation Room activation.

Step 6 — Calibrate: Historical analysis + model validation + expert elicitation + pilot testing. Output: Calibration Report.

Step 7 — Governance: Technical Working Group proposes; regional thresholds are formally agreed and endorsed by Member States through the SHOC Governing Board; Member States refine nationally to suit local context. This governance structure ensures that regional coordination mechanisms such as SHOC intervene at clearly defined and regionally mandated levels.

Step 8 — Maintenance: Quarterly operational review. Annual calibration. Post-activation AAR.

7.7 Practical Examples

Flood Tier 3: Rainfall > 90th percentile AND soil saturation high AND population in flood footprint > X (from Situation Room GIS exposure layer) OR critical infrastructure exposed.

Cholera Tier 3: Case incidence > baseline AND rate suggests expansion AND health capacity breached (bed occupancy from Situation Room vulnerability dashboard).

8. SOP Multi-Hazard Anticipatory Action Coordination

> ⚠ **CRITICAL:** SHOC coordinates and advises; deployment of physical assets is the sovereign responsibility of Member States. Physical deployment and supply chain operations are governed by SOP SCM (Logistics and Supply Chain), not this SOP.

A true multi-hazard approach addresses compounding effects. Thresholds capture cascading impacts. The Situation Room's GIS-derived impact and exposure estimates provide the spatial evidence base for anticipatory action triggering and coordination with NDMAs.

Situation Room Integration: Anticipatory action triggers are monitored on the Situation Room Multi-Hazard Risk Dashboard. When composite risk scores (hazard × exposure × vulnerability/capacity) approach activation tiers, the Situation Room generates threshold breach alerts and impact estimates that inform the decision authority chain and NDMA coordination meetings.

8.1 Relevant Annexes

Annex 2.4-A: Anticipatory Action Trigger Protocol [IFRC FbF Manual; OCHA AA]

Annex 2.4-B: EAP Activation Checklist [OCHA AA Framework]

Annex 2.4-C: Resource Pre-Positioning Request Form [UNHRD Partner Handbook]

Annex 2.4-D: After-Action Review Template [IFRC FbA Learning Framework]

8.2 Process Flow Diagram

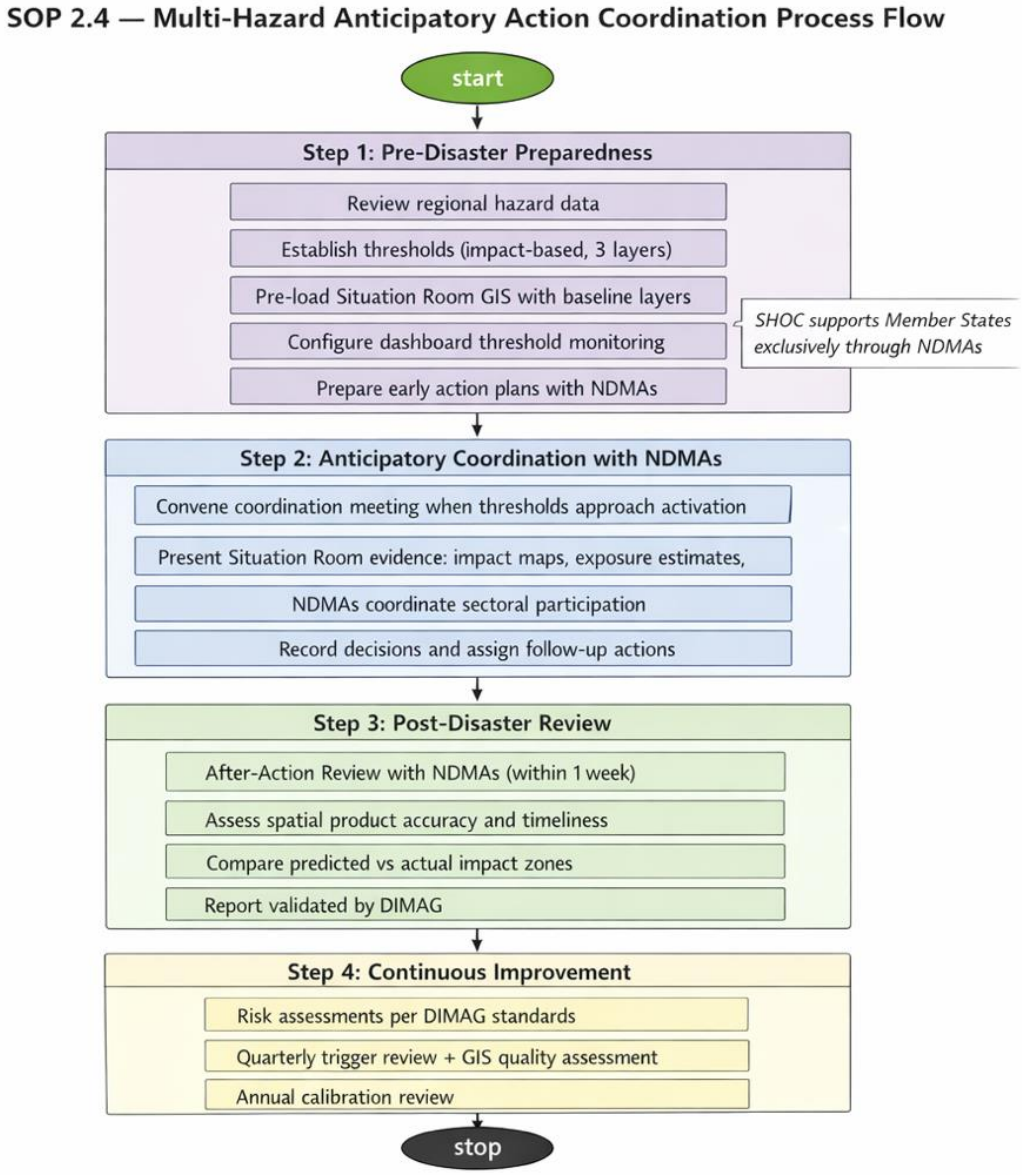

Figure 8: SOP MULTI HAC — Multi-Hazard Anticipatory Action Coordination

Step 1: Pre-Disaster Preparedness

Review regional hazard data. Establish thresholds using impact-based methodology (3 layers). Pre-load Situation Room GIS with baseline exposure and vulnerability layers. Configure threshold monitoring on Multi-Hazard Risk Dashboard. Prepare early action plans with NDMAs.

Step 2: Anticipatory Coordination with NDMAs

> **GOVERNANCE NOTE:** SHOC supports Member States primarily through NDMAs as the designated entry point, with direct engagement of other relevant national agencies in alignment with NDMAs.

SHOC convenes anticipatory coordination meetings with NDMAs when: thresholds approach activation; multi-country impacts are plausible; joint interpretation is required; early action requires national coordination.

8.2.1 Coordination Meeting Protocol

Step	Activity	Detail

1	Meeting Convening	Head EW&M convenes meeting when any threshold approaches activation tier. Invitations issued via NDMA primary channels. NDMAs coordinate sectoral participation per national arrangements.
2	Evidence Presentation	Situation Room spatial products presented as shared evidence base: impact maps, exposure estimates, risk composites, resource pre-positioning maps. All products are GIS-derived from Layer 2 analysis.
3	Shared Risk Interpretation	NDMAs and SHOC jointly interpret risk data. NDMAs provide ground-truth verification. Discrepancies documented and resolved per Section 1.9 protocol.
4	Action Planning	Agree proportionate anticipatory actions per Member State context. Identify resource pre-positioning requirements. Assign follow-up responsibilities with deadlines.
5	Decision Recording	All decisions recorded using the Coordination Meeting Decision Record template. Action items assigned with responsible parties and deadlines. Record circulated within 4 hours.
6	Follow-up Tracking	Action items tracked on NDMA Coordination Status Board. Follow-up verification at next meeting or within agreed timeline.

Step 3: Post-Disaster Review

After-Action Review with NDMAs within 1 week. Include assessment of Situation Room spatial product accuracy and timeliness. Post-event spatial analysis comparing predicted vs actual impact zones. Report validated by DIMAG.

Step 4: Risk Management and Continuous Improvement

Risk assessments using DIMAG standards. Quarterly trigger review including GIS data quality assessment. Annual calibration review including Situation Room system performance.

8.3 After-Hours Protocol

Given that Full Activation requires 24/7 operations, the following after-hours protocol applies:

Step	Protocol	Detail
1	Duty Officer Designation	A rotating Duty Officer roster is maintained for after-hours, weekends, and public holidays. The Duty Officer has authority to initiate Heightened Monitoring and convene the on-call team.
2	Activation Notification	Duty Officer notifies Head EW&M via designated emergency communication channel. Head EW&M activates the escalation chain per the Decision Authority Matrix.
3	Minimum Response Time	Duty Officer: immediate. Head EW&M: ASAP (target: 30 minutes). Full on-call team: ASAP (target: 2 hours).
4	Standby Arrangements	During elevated hazard seasons, pre-designated staff maintain standby status with remote access to Situation Room dashboards enabled.

9. Enhanced Risk Analysis Matrix

The following risk analysis matrix has been updated to include new risks identified through the gap analysis and to reflect the enhanced mitigation controls provided by the six-phase lifecycle model. New entries are marked with an asterisk (*).

#	Risk Category	Description	L	I	Inher.	Resid.	Mitigation / Control
1	Infrastructure Failure	Hardware or antenna malfunction during deployment	3	4	12	6	Spare routers & VSATs; daily diagnostics; SHOC NOC remote

							support; triple-redundancy Layer 1-3.
2	Power Outage	Loss of power to VSAT hub or switches	3	3	9	4	Hybrid solar/generator; battery banks (≥24h); UPS; OSSOC Go-Kit portable power.
3	Cyber Intrusion	Malware attack on VPN or cloud nodes	2	5	10	4	Integrated SOC-NOC monitoring; 2FA; firewall; automated threat-based failover.
4	Data Loss	Backup failure or human error	2	5	10	4	Hourly cloud backups; restricted edit rights; version control; RPO=0 replication.
5	Frequency Licence Delay	Administrative hold-up in radio licensing	2	5	10	4	Pre-Clearance Cache; ITU/SADC MOUs; pre-licensed VSAT; Alert-phase activation.
6*	OSSOC Connectivity Failure	OSSOC unable to establish comms within 8h	2	5	10	4	Dedicated Go-Kit; triple-redundancy; pre-configured VPN; ICT Specialist in ERT.
7*	EW Feed-Back Loop Failure	Situation Room unable to relay hazard updates to OSSOC	2	4	8	3	Automated data push; voice briefings 2x daily; emergency SMS; multiple paths.
8*	Handoff Accountability Gap	Unclear telecom responsibility during transit	3	3	9	3	Formal Handoff Protocol; Accountability Transfer Form; SHOC NOC oversight.
9	Personnel Safety	Hazardous conditions during deployment	3	4	12	6	Daily safety briefings; PPE; IFRC §17; medical evac plan.
10	Logistics Delay	Customs or transport bottlenecks	3	3	9	5	Customs waivers; WFP MoU; advance shipping docs; Go-Kit air-transport ready.
11	Stakeholder Coordination	Poor data sharing SHOC-NDMA	2	3	6	3	Weekly joint ops calls; CAP protocol; Situation Room briefs; EW feed-back loop.
12	Environmental Hazard	Storm damage at field site	3	4	12	6	Elevated masts; weather-proof enclosures; hazard-to-telecom profiles.
13*	ETK Readiness Degradation	Pre-positioned ETK batteries/firmware degrade	3	3	9	3	Automated ETK Status Monitoring System; dashboard alerts; 8h/24h resolution.

10. Enhanced Performance Monitoring Indicators (PMIs)

Indicator	Definition	Target	Frequency	Responsible
OSSOC Connectivity Speed	Time from ERT arrival to OSSOC full operational connectivity	≤ 8 hours	Per operation	ICT Specialist (ERT)
Deployment Speed	Time from activation to full SHOC-NDMA operational setup	≤ 48 hours	Per operation	ICT Deployment Team
Network Uptime	SHOC backbone infrastructure availability	≥ 99.9%	Continuous	NOC / ICT Office
Auto-Failover Speed	Time for automated failover between redundancy layers	≤ 30 seconds	Per failover event	NOC Team
Data Integrity (RPO)	Critical data loss events	0 events (RPO=0)	Continuous	Cloud Admin / IM Officer

System Redundancy	Active independent backup channels across 3 layers	≥ 3 layers	Quarterly	ICT Infrastructure
ETK Readiness Rate	Pre-positioned ETKs passing all status checks	≥ 95%	Monthly	ICT Specialist / Warehouse
EW Feed-Back Loop Latency	Time for hazard alert to reach OSSOC from Sit Room	≤ 5 minutes	Per event	Situation Room / ICT Specialist
Cross-Regional Interoperability	Successful data exchange with AUC	100%	Quarterly	SHOC Secretariat
Drill Completion Rate	Quarterly readiness drills completed vs planned	100%	Quarterly	ICT Manager / MEL

— END OF SOPs —

All Annexes, Annex Master Index, Glossary, Document Control, and References are contained in the companion document: SOP Annexes and Ancillaries